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Section: **D – (D1)**

EXP: 3

Aim: **Consider the problem definition in practical 1. Write and execute SQL functions- aggregate, numeric, date, string, and conversion for the following queries:**

a) Find the average salary of instructors in the Computer Science department.

⇨ **select d.name, avg(i.salary) as AVG_SAL from instructor i join department d on i.dept_id = d.dept_id where d.name = 'Computer Science' group by d.name;**

b) Find the average salary in each department.

⇨ **select d.dept_id, d.name, avg(i.salary) as AVG_SAL from instructor i join department d on i.dept_id = d.dept_id group by d.dept_id, d.name;**

c) What is the total salary of all teachers earning more than 30K?

⇨ **select sum(salary) as total_sal from instructor where salary > 30000;**

d) List the number of students enrolled in each course

⇨ **select C.course_id, C.title, COUNT(E.student_id) as enrolled_students from course C left join section s on C.course_id = S.course_id left join enrollment E on S.section_id = E.section_id group by C.course_id, C.title order by C.course_id;**

e) Find the names of all departments whose building name include the substring“Symbio”.

⇨ **select name from department where lower(building) LIKE '%symbio%';**

f) Find the names of the departments whose names contain the string “sci” as asubstring, regardless the case.

⇨ **select name from department where lower(name) LIKE '%sci%';**

g) Display the length of the name of all the students.

⇨ **select name, length(name) AS name_length FROM student;**

h) Retrieve total number of instructors.

⇨ **select count(*) AS total_instructors FROM instructor;**

i) Add hiredate column in the instructor table. Insert 4 rows in the table. Displaythe names of instructor who are working for the past 5 years.

- **alter table instructor add hiredate date;**
- **UPDATE instructor SET hiredate = DATE '2021-01-30' WHERE instructor_id = 1;**
- **UPDATE instructor SET hiredate = DATE '2020-04-13' WHERE instructor_id = 2;**
- **UPDATE instructor SET hiredate = DATE '2019-07-21' WHERE instructor_id = 3;**
- **UPDATE instructor SET hiredate = DATE '2018-11-05' WHERE instructor_id = 4;**
- **UPDATE instructor SET hiredate = DATE '2022-02-14' WHERE instructor_id = 5;**
- **UPDATE instructor SET hiredate = DATE '2020-09-30' WHERE instructor_id = 6;**
- **UPDATE instructor SET hiredate = DATE '2017-06-18' WHERE instructor_id = 7;**
- **select * from instructor where hiredate < add_months(sysdate, -60);**

j) Display the list of instructor who have joined after 1st Jan 2021 or before 30st Jun 2021.

- **SELECT * FROM Instructor where hiredate BETWEEN DATE '2021-01-01' and date '2019-12-31';**

k) Display name of those instructors who are going to retire 01-Jan-24. If themaximum job is period is 3 years.

- **select * FROM Instructor WHERE HireDate <= date '2021-01-01';**
- **select * from instructor where hiredate = add_months(sysdate, -60) and hiredate < date '2026-01-01';**

l) Display those instructors whose salary contains at least 4 digits and salary above 75000.

- **select name, salary from instructor where salary >= 1000 and salary > 75000;**

m) Print a list of instructors displaying 'Less Salary' if less than 60000 if exactly 60000 display as 'Exact Salary' and if greater than 60000 display 'More Salary'.

- **select name, case when salary < 60000 then 'less salary' when salary > 60000 then 'More salary' end as salary_category from instructor;**

n) Display names of all the instructors concatenated with their department names.

- **select i.name || ' => ' || d.name from instructor i join department d on i.dept_id=d.dept_id;**

o) Retrieve distinct number of instructors.

- **select d.name, count(i.name) from instructor i join department d on i.dept_id = d.dept_id group by d.name;**

p) Find the names of all instructors whose salary is greater than at least one instructor in the Biology department.

- **SELECT * FROM instructor WHERE salary > ANY (SELECT salary FROM instructor I JOIN department d ON i.dept_id = d.dept_id WHERE d.name = 'Biology');**